Statistics Final Project Report

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For this project, I was curious about whether murder rates were associated with different regions across the United States. In other words, I wanted to know if living in a specific region is associated with higher or lower numbers of murder while accounting for differences in state populations. To answer this question, I used the "murders" dataset from the built-in dslabs package in RStudio programming, which included variables for state, region, population, and total murders that have occurred in each state. I was able to use this dataset to explore my question by running a linear regression frequentist model using total murders from each state as my dependent, or outcome, variable, and region and population as independent variables. From running a linear regression, I was able to determine if there was a relationship between total murders and regions across the United States while controlling for states population. I also explored descriptive statistics of each variable to understand more about the data and created visualizations to better understand distributions of total murders.

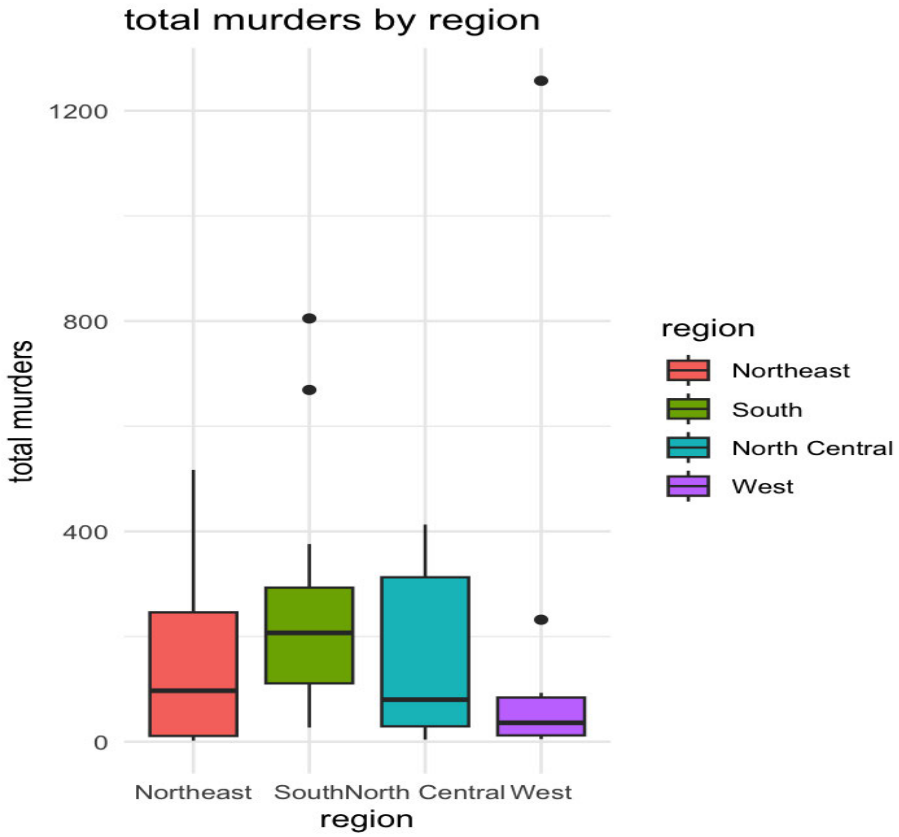
Before examining my data and running analyses, I predicted what I thought the outcome of my research question would be. When thinking of where murders in the United States typically take place, I thought of large cities being hot zones for violent crime. Large cities also have higher prevalence rates of gang members who are commonly involved in violent crimes, aiding to increased murder rates. When looking at a map of the United States split up by region, I saw how certain regions, specifically the South and West, have more cities and populated areas that could contribute to higher murder rates. Other regions, such as the Midwest, were less populated and have a geographic landscape that doesn't allow for as many cities, potentially limiting murder totals. Therefore, I predicted that the South, being the most populated, would be

associated with an increased murder total followed by the West, and Midwestern regions would have lower murder totals due to smaller population. By using population as a predictor in my linear regression, I was able to examine whether differences in murder totals were truly associated with region, or if state population was the driving factor.

The dataset revealed 51 U.S. states, including the District of Columbia, with variables for total murders, state population, and the region of the U.S. that the state is apart of. The total number of murders per state ranged from two to 1,257, suggesting that the distribution across states is very uneven. Descriptive statistics of the variables showed that the mean, or average, number of murders per state was 184.37. However, this mean is misleading with the median showing an average of 97. This suggests that the distribution is skewed with few states having high rates of murder compared to the rest of the country. The standard deviation for total murders is 236.13, with the large number further suggesting there is a lot of variability in murders across states. There is also a large distribution across population per state, ranging from 563,626 to 37,253,956. The mean is again misleading showing a population average of 6,075,769, where the median population is 4,339,367. The median is more reliable for understanding average state population where the mean is skewed by a few states with very large populations. The population standard deviation is 6,860,669, indicating variability across state populations. Running a frequency table, I was able to see how many states were apart of each region. The regions in the dataset included Northeast, South, North Central, and West. The distribution across regions was not even with the South having the most states (17), followed by the West (13), North Central (12), and lastly Northeast (9). When comparing total murders and population across regions, descriptive statistics again revealed notable differences. The median total murders per state were highest in the South (207), followed by the Northeast (97), North Central (80), and West (36).

These were much lower than the mean averages, suggesting few states had high murder rates. Median population per state also varied by region with North Central at the highest (5,495,456) and the West with the lowest (2,700,551). This suggests that population may not be the only factor impacting murder rates across states, but the region of the state could also influence murders. Figure 1 is a boxplot of total murders by region with each box representing murders from a specific region. Inside the box, the black line represents the median number of murders and the whole box spans from the first quartile to the third quartile. The whiskers extending from the box represent minimum and maximum murders in each region and dots above the box are outliers showing skewed data points. From the figure we can see Southern regions have the greatest number of murders on average where the West has the least amount. However, Western regions have the greatest variability due to the extremely high outlier of one state having an unusually high total number of murders compared to other states in the West.

Figure 1: Total Murders by Region



To examine whether state murder rates differed by region after controlling for population, I ran a multiple linear regression with total murders as my outcome variable and region and population as predictors. The regression question for the model was Total Murders ~ Region + Population. This model revealed statistically significant results for population (β=3.29x10^-5, p<.001) suggesting for every one person increase in state population, murders increase on average by 3.29e-05. Population was therefore a positive predictor with every 100,000 person increase in a state's population, total murders increased by about 3.3. Holding population constant, the only region that significantly differed from the Northeast was the South (β=6.19e^+1, p<0.015) suggesting that states in the South experienced 62 more murders on average.

Overall, these results show that even after controlling for population, region was still associated with murder rates. However, this effect was only significant in the South and no other regions significantly differed from the Northeast. This suggests that Southern states experience more total murders than would be expected only based on population, indicating regional factors associated as being Southern may increase total murders. While my prediction for Southern states to have higher murder totals was supported from the results, my prediction for Western states was not supported. I had originally thought the West would have the second highest total murders, but they actually had the lowest showing that factors other than population impact murder totals. This made me curious about which factors specifically in the South contribute to higher murder totals. Since gun ownership is higher in Southern states, I would next want to study how this factor could impact total murder in the South.